

View Review

Paper ID

830

Paper Title

Real-Time Detection of Financial Crimes Using an AI-Driven Hybrid Model of Behavioural and Transactional Patterns

REVIEW QUESTIONS

1. Comments to Authors

The provided paper introduces a new and properly structured process of real-time detection of financial crimes involving the creation of an AI-based hybrid system, which combines

behavioral and transactional data analysis. This combination of output of an LSTM-based behavioral module and an XGBoost/Isolation Forest-based transactional module through a fusion layer is a great strength and it can take care of the drawback of traditional systems being delayed

to give reaction and false-positive.

The performance measures of the model are remarkable and the model shows accuracy of 99.1%, precision of 98.7%, recall of 98.2, and AUC-ROC of 0.996 and this outperforms a few baseline models. Of note among the critical contributions, the focus is on a real-time processing pipeline build using Apache Kafka and Apache Flink, which helps enable less than 100 milliseconds of low-latency inference, bringing the system to high-volume financial environments in practice.

To further strengthen the paper, I have a few suggestions for a minor revision:

☐ The paper indicates that synthetic datasets combining Kaggle datasets was used, but more specific description in terms of size and other features would add to the reproducibility

and credibility of the study. As an example, it would be advantageous to introduce the number of transactions (or the period of coverage).

☐ It would be a good idea to either present an ablation study or perform a closer examination of the individual performance of the behavioral (LSTM) and transactional (XGBoost/Isolation Forest) modules once again on the overall performance. Although the paper claims that the fusion approach performed significantly better than the application of either of the models alone, simply reporting the results of the two models independently would prove this hypothesis to be true.

☐ In Figure 1, the flowchart refers that the use of the CNN - LSTM + Anomaly Detection, yet the textography is more about LSTM, XGBoost and Isolation Forest. It would also be good clarifying whether the CNN component has a role in the hybrid model or not and, therefore, add consistency and clarity. The application of CNN-RNN has also been cited in relationship to other researches.